SHREWD IOT CENTRED INSTANT ASTHMA DUE TO DUST RISK PREDICTOR

A PROJECT REPORT

Submitted by

ABHILASH A S (962822205304)

ANTO JEBA LISPEN A (962822205026)

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

UNIVERSITY COLLEGE OF ENGINEERING NAGERCOIL

ANNA UNIVERSITY :: CHENNAI 600 025

MAY 2026

ANNA UNIVERSITY: CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report **“Shrewd IoT centred Instant Asthma due to Dust Risk predictor”** is the Bonafide work of **“Abhilash A S (962822205304), Anto Jeba Lispen A (962822205026)”** who carried out the project work under my supervision.

Dr. K.L. NEELA, M.E., Ph.D.,	**Dr J Banumathi B.Tech.,M.E, Ph.D.,**
HEAD OF THE DEPARTMENT,	**MENTOR,**
Assistant professor (Sl. Gr),	Assistant Professor (Sl. Gr.),
Computer Science Engineering,	Computer Science Engineering,
University College of Engineering,	University College of Engineering,
Nagercoil, Konam – 629 004	Nagercoil, Konam – 629 004

PROJECT TEAM

Abhilash A.S
Anto Jeba Lispen A

Academic Year : 2025-2026

ABSTRACT

Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor is an intelligent healthcare monitoring system developed to detect and monitor asthma risk conditions caused by dust exposure in real time using IoT technology. The proposed system integrates ESP32 microcontroller, dust sensing technology, biomedical sensors, and Raspberry Pi for continuous environmental and health monitoring. The system is designed to monitor important parameters such as dust concentration, oxygen saturation level (SpO2), heart rate, and body temperature to identify possible asthma risk conditions instantly. The collected sensor data is processed by the ESP32 and transmitted wirelessly through Wi-Fi communication to the Raspberry Pi monitoring station for real-time visualization and analysis. The developed system provides a portable, low-cost, and user-friendly solution suitable for hospitals, industries, polluted environments, and home healthcare applications. The proposed project helps in early detection of asthma-related risks, improves patient safety, and supports continuous health observation through smart IoT-based monitoring technology while enabling future enhancements such as cloud storage, emergency alerts, mobile applications, and AI-based health prediction systems.

ACKNOWLEDGEMENT

First and foremost, we thank **GOD** for his grace to enable us this project work successfully.

Words cannot be expressed to show our gratitude to our respected DEAN **Dr. V. A. NAGARAJAN, Ph.D**., for his support and freedom throughout our course.

We sincerely thank our Head of the Department, **Dr. K.L. NEELA, M.E., Ph.D.,** Assistant Professor (Sl.Gr.), Department of Computer Science and Engineering and Head of the Department, **Dr. A. RADHAKRISHNAN, M.E., Ph.D.,** Assistant Professor (Sl.Gr.), Department of Information Technology, for their valuable guidance and continuous support throughout the project.

We thank our PROJECT MENTOR **Dr J BANUMATHI., B.Tech.,M.E, Ph.D**., Assistant Professor (Sl. Gr.), Department of Computer Science Engineering, for providing us with the valuable suggestions and constant encouragement towards our project.

And our sincere thanks to all **Department Teaching and Non – Teaching Staff** members for their constant support for making this project a grand success.

Finally, we wish to thank Our Parents for their blessings to complete the project successfully

TABLE OF CONTENTS

LIST OF ABBREVIATION

ABBREVIATION	EXPANSION
IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller
BPM	Beats Per Minute
SpO2	Peripheral Oxygen Saturation
TFT	Thin Film Transistor
Wi-Fi	Wireless Fidelity
MQTT	Message Queuing Telemetry Transport
ECG	Electrocardiogram
AI	Artificial Intelligence
GPIO	General Purpose Input Output
I2C	Inter-Integrated Circuit
HTTP	HyperText Transfer Protocol
USB	Universal Serial Bus
AQI	Air Quality Index
IR	Infrared
CPU	Central Processing Unit
IDE	Integrated Development Environment
OS	Operating System
TP4056	Lithium Battery Charging Controller Module
LCD	Liquid Crystal Display
PCB	Printed Circuit Board
DC	Direct Current
LED	Light Emitting Diode
IP	Internet Protocol
GUI	Graphical User Interface

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The introduction for our project titled "Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor" focuses on developing a smart healthcare and dust risk prediction monitoring system for real-time observation of user vital signs and environmental dust conditions.

In today's healthcare environment, continuous monitoring of users is very important for detecting abnormal health conditions and dust-related asthma risk at an early stage. Traditional monitoring systems require manual observation and are not always suitable for continuous remote monitoring.

To overcome these limitations, we designed and developed an IoT-based portable monitoring system using ESP32 and Raspberry Pi technologies. The proposed system is capable of monitoring important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and surrounding dust concentration.

The MAX30100 sensor is used for measuring heart rate and SpO2, while the DS18B20 sensor is used for measuring body temperature. Dust sensing technology is additionally used for predicting asthma risk conditions caused by harmful dust exposure.

In our project, the ESP32 microcontroller collects data from the sensors, processes the readings, predicts possible asthma risk conditions, and transmits the information wirelessly through Wi-Fi communication.

The Raspberry Pi acts as the monitoring station that receives the sensor and dust prediction data and displays the live values on a TFT display screen. This enables continuous observation of user health conditions and environmental safety in real time.

The portable nature of the ESP32 unit powered using a rechargeable battery system makes the device compact and easy to use for healthcare and dust monitoring applications. The system is designed to provide low-cost, wireless, and efficient user monitoring and asthma prediction support for hospitals, home healthcare, industrial environments, and remote medical applications.

Our project mainly aims to improve user monitoring by providing real-time data visualization and wireless communication using IoT technology. Future enhancements of the system may include cloud connectivity, mobile application support, emergency alert systems, dust analytics, and AI-based asthma risk prediction for advanced healthcare solutions.

1.2 OBJECTIVE

The main objective of our project is to develop a portable and real-time healthcare and dust risk prediction monitoring system using IoT technology for continuous observation of user vital signs and environmental dust levels. The system is designed to monitor important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and dust concentration using sensors connected to the ESP32 microcontroller. Another objective of the project is to transmit the collected data wirelessly to a Raspberry Pi monitoring station through Wi-Fi communication for live monitoring and visualization. Our project also aims to provide a low-cost, compact, and easy-to-use healthcare solution that can be used in hospitals, home healthcare, industrial safety monitoring, and remote user monitoring applications. In addition, the system is developed to support future enhancements such as cloud connectivity, mobile application integration, emergency alert systems, dust prediction analytics, and advanced health analysis for improved healthcare services.

1.3 PROBLEM STATEMENT

In many healthcare environments, continuous monitoring of user vital signs is still performed manually, which can be time-consuming and less effective for real-time observation. Users suffering from respiratory problems are more vulnerable to dust exposure and require regular monitoring of parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and surrounding dust conditions. Traditional monitoring systems are often expensive, bulky, and not suitable for portable or remote healthcare applications. In rural and remote areas, lack of continuous monitoring facilities can delay the detection of abnormal health conditions and dust-related asthma risks, affecting user safety.

To overcome these problems, there is a need for a low-cost, portable, and wireless healthcare monitoring system that can continuously monitor vital signs and predict asthma risk due to dust exposure in real time and display the data effectively. Therefore, our project proposes a Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor using ESP32 and Raspberry Pi to provide continuous user monitoring, wireless data transmission, dust risk prediction, and real-time visualization of health parameters for improved healthcare support.

1.4 PURPOSE

The purpose of our project is to develop a smart and portable healthcare and dust prediction monitoring system that can continuously monitor user vital signs and environmental dust levels in real time using IoT technology. The system is designed to measure important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and dust concentration using sensors connected to the ESP32 microcontroller. The collected data is transmitted wirelessly to the Raspberry Pi monitoring station for live visualization, asthma risk prediction, and continuous observation.

The main purpose of the project is to provide a low-cost, wireless, and easy-to-use healthcare solution for hospitals, home healthcare, industrial safety monitoring, and remote user monitoring applications. The project also aims to reduce manual monitoring efforts and help in the early detection of abnormal health conditions and dust-related asthma risks through continuous real-time monitoring. Additionally, the system creates a foundation for future healthcare technologies such as cloud-based monitoring, emergency alert systems, dust analytics, and AI-based asthma risk prediction

1.5 SCOPE

The scope of our project is to develop a portable and IoT-based healthcare and dust risk prediction monitoring system for real-time observation of user vital signs and surrounding environmental dust conditions. The system is designed to monitor important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and dust concentration using ESP32 and Raspberry Pi technologies. The project mainly focuses on wireless communication, real-time data processing, asthma risk prediction, and live visualization of health and dust data on a TFT display.

The proposed system can be used in hospitals, home healthcare, clinics, industries, and remote user monitoring applications where continuous observation of user health and environmental safety is required. Since the system is portable and rechargeable, it can also be used for personal healthcare monitoring and emergency situations in polluted or dust-prone environments. The project provides a low-cost and efficient solution compared to traditional monitoring systems.

The future scope of the project includes integration with cloud platforms for online data storage, mobile application support for remote access, emergency alert systems for critical conditions, dust analytics for air quality monitoring, GPS tracking for user location monitoring, and AI-based asthma risk prediction for detecting abnormal respiratory conditions.

1.6 CONTRIBUTION OF THE PROJECT

The contribution of our project is the development of a low-cost, portable, and IoT-based healthcare and dust risk prediction monitoring system for real-time user observation. The project contributes to the healthcare field by providing continuous monitoring of vital signs such as heart rate, oxygen saturation level (SpO2), body temperature, and surrounding dust levels using ESP32 and Raspberry Pi technologies. The system enables wireless communication and real-time data visualization, which helps in reducing manual monitoring efforts and improving user care and respiratory safety.

Another important contribution of the project is the integration of embedded systems, IoT communication, dust monitoring, and real-time asthma risk prediction into a single compact device. The portable design of the system makes it suitable for hospitals, home healthcare, industries, and remote user monitoring applications. The project also creates a foundation for future enhancements such as cloud connectivity, mobile application support, emergency alert systems, dust analytics, and AI-based asthma prediction technologies, which can further improve smart healthcare and environmental monitoring systems.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The literature review focuses on the study of existing IoT-based healthcare monitoring systems developed for real-time user monitoring and remote healthcare applications. Various research papers discuss the use of ESP32 microcontrollers, biomedical sensors, wireless communication, and real-time data visualization for monitoring vital signs such as heart rate, oxygen saturation level (SpO2), and body temperature.

The review of existing studies helps in understanding different technologies, system architectures, communication methods, advantages, and limitations of current healthcare monitoring systems. It also provides knowledge for developing an efficient, low-cost, portable, and real-time healthcare monitoring solution using ESP32 and Raspberry Pi.

[1] Hemaxi Vyas et al – "A Portable IoT-Based Health Monitoring Framework Using ESP32 for Isolated and Home-Based User Care" (2025)

This paper presents the development of an IoT-based health monitoring system using the ESP32 microcontroller for real-time user monitoring. The system is designed to measure important health parameters such as heart rate, oxygen saturation level (SpO2), and body temperature using biomedical sensors. The collected sensor data is processed and displayed through a web interface and mobile application for continuous health observation.

The system mainly focuses on providing a low-cost, portable, and wireless healthcare monitoring solution suitable for hospitals, home healthcare, and remote user monitoring applications. The results showed that the system can continuously monitor user vital signs in real time and provide quick medical alerts during abnormal health conditions. The study also highlights future improvements such as cloud connectivity, AI-based analysis, and additional sensor integration for advanced healthcare monitoring.

[2] Himanshi Manjhi – "IoT Based User Health Monitoring System Using ESP32 and Blynk App" (2025)

This paper presents an IoT-based user health monitoring system developed using the ESP32 microcontroller and Blynk IoT platform for real-time remote healthcare monitoring. The system is designed to monitor important health parameters such as heart rate, body temperature, and oxygen saturation level (SpO2) using biomedical sensors. The collected sensor data is transmitted wirelessly through Wi-Fi communication to the Blynk cloud platform, allowing healthcare professionals to monitor user health remotely through a mobile application.

The system provides real-time data visualization and emergency alert notifications when abnormal health conditions are detected. The study mainly focuses on developing a low-cost, portable, and cloud-based healthcare monitoring solution suitable for elderly users, chronic disease monitoring, and remote healthcare applications. The results showed reliable sensor performance, stable wireless communication, and effective real-time monitoring capabilities for continuous user observation.

[3] Jannatun Ferdous et al – "Implementation of IoT Based User Health Monitoring System Using ESP32 Web Server" (2023)

This paper presents the implementation of an IoT-based user health monitoring system using the ESP32 web server for real-time monitoring of user vital signs. The system is designed to measure important health parameters such as pulse rate, body temperature, and oxygen saturation level (SpO2) using biomedical sensors including MAX30100, DS18B20, and DHT11 sensors. The collected sensor data is transmitted wirelessly through Wi-Fi communication and stored on a web server for remote monitoring and analysis. The proposed system allows doctors and healthcare professionals to monitor user health conditions anytime and anywhere using smart devices and web applications. The study mainly focuses on providing a flexible, low-cost, wireless, and efficient healthcare monitoring solution for remote user care. The results showed that the system successfully monitored and displayed real-time user health parameters through the ESP32 web server and mobile devices for continuous healthcare observation.

[4] Yusra M. Obeidat et al – "An Embedded System Based on Raspberry Pi for Effective Electrocardiogram Monitoring" (2023)

This paper presents the development of an embedded system based on Raspberry Pi for effective real-time electrocardiogram (ECG) monitoring in healthcare applications. The proposed system is designed to monitor ECG signals accurately using embedded hardware and signal processing techniques. The system includes analog and digital electronic circuits for signal amplification, noise filtering, data conversion, and real-time visualization of ECG signals through a graphical display. The Raspberry Pi is used as the main processing unit for monitoring, storing, and displaying ECG data efficiently.

The study mainly focuses on developing a portable, low-cost, and user-friendly healthcare monitoring device capable of detecting abnormal heart conditions in real time. The proposed system successfully measured ECG signals at different heart rates and displayed important ECG peaks required for health condition analysis. The system also supports wireless monitoring and cloud-based data storage for continuous user observation and future medical diagnosis. The results showed that the Raspberry Pi-based monitoring system provides efficient real-time healthcare monitoring with improved signal quality, portability, and accuracy for medical applications

[5] Suliman Abdulmalek et al – "IoT-Based Healthcare-Monitoring System towards Improving Quality of Life: A Review" (2022)

This review paper discusses the role of Internet of Things (IoT) technology in modern healthcare-monitoring systems and how it improves the quality of human life. The paper explains that IoT-based healthcare systems use smart sensors, wearable devices, wireless communication, cloud computing, and real-time monitoring to observe users remotely and provide medical assistance efficiently. These systems are useful for monitoring vital health parameters such as heart rate, blood pressure, body temperature, ECG, oxygen saturation, and other physiological signals. The study also highlights applications such as remote healthcare, preventive healthcare, emergency monitoring, and wearable health devices.

The authors reviewed many existing IoT healthcare-monitoring systems and compared their methodologies, hardware technologies, communication protocols, features, evaluation metrics, and limitations. Common technologies discussed in the paper include Arduino, Raspberry Pi, NodeMCU, Bluetooth, Wi-Fi, BLE, Zigbee, LoRa, cloud platforms, and wearable sensors. The paper explains how IoT enables real-time data collection, storage, analysis, and remote access for doctors and users. It also discusses the Internet of Wearable Things (IoWT), wireless sensor networks, and cloud-based medical monitoring systems that improve healthcare accessibility and reduce medical costs.

In addition, the review identifies several challenges and open issues in IoT healthcare systems, including data security, privacy protection, energy consumption, network reliability, integration complexity, and quality of service. The paper concludes that IoT-based healthcare systems have huge potential for future smart healthcare applications because they provide continuous user monitoring, early disease detection, and faster medical response. The authors recommend improving system security, interoperability, sensor accuracy, and cloud integration for future healthcare technologies.

CHAPTER 3

SYSTEM ANALYSIS

3.1 INTRODUCTION

The system analysis of our project focuses on the design and implementation of an IoT-based real-time healthcare and dust risk prediction monitoring system using ESP32 and Raspberry Pi.

The proposed system is developed to continuously monitor important user vital signs such as heart rate, oxygen saturation level (SpO2), body temperature, and surrounding dust concentration in real time.

The analysis of the system includes hardware components, software requirements, wireless communication, data processing, asthma risk prediction, and real-time visualization of health parameters and environmental conditions.

In our project, the MAX30100 sensor is used for measuring heart rate and SpO2, while the DS18B20 sensor is used for measuring body temperature. Dust sensing technology is additionally used for monitoring environmental dust levels and predicting asthma risk conditions.

These sensors are connected to the ESP32 microcontroller, which acts as the main processing and communication unit. The ESP32 collects the sensor readings, processes the data, predicts possible asthma-related risks, and transmits the information wirelessly through Wi-Fi communication.

The Raspberry Pi acts as the monitoring station that receives the data sent by the ESP32. The received information is processed and displayed on a TFT display for continuous observation of user health conditions and environmental dust status. The system is designed to provide portable, wireless, and real-time monitoring with low power consumption and easy operation. The overall system analysis helps in understanding the working principle, data flow, hardware integration, and communication process involved in the project. It also ensures that the system can provide accurate monitoring, reliable wireless communication, efficient asthma risk prediction, and effective real-time healthcare visualization.

3.2 PROBLEM STATEMENT

In the current healthcare environment, continuous monitoring of user vital signs is an important requirement for detecting health abnormalities at an early stage.

Traditional monitoring methods mainly depend on manual observation by medical staff, which may not provide continuous real-time monitoring and can increase the workload in hospitals and healthcare centers. Users with respiratory problems and dust allergies require regular monitoring of parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and surrounding dust exposure levels.

Existing healthcare monitoring devices are often expensive, bulky, and difficult to use for portable or remote monitoring applications. In rural and remote areas, lack of proper monitoring systems may delay medical attention and affect user safety. Continuous monitoring of dust concentration and asthma risk conditions is also difficult using traditional healthcare systems.

There is a need for a compact, low-cost, wireless, and real-time healthcare and dust risk prediction monitoring solution that can continuously monitor user vital signs and environmental dust levels while providing easy visualization of health data and instant asthma risk prediction.

To solve these issues, our project proposes a Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor using ESP32 and Raspberry Pi for wireless monitoring, real-time data processing, dust risk analysis, and continuous visualization of user health conditions and environmental safety.

3.3 EXISTING SYSTEM

The existing healthcare monitoring systems mainly rely on traditional medical devices that require manual operation and direct supervision by healthcare professionals. Most conventional systems are designed for hospital environments and are not suitable for portable or remote user monitoring applications. These systems often involve high-cost equipment and complex monitoring setups.

In many cases, user vital signs such as heart rate, oxygen saturation level, body temperature, and environmental dust exposure are measured separately using different devices.

Continuous monitoring is difficult because the data is not always transmitted wirelessly or monitored in real time. Some systems also lack portability, making them unsuitable for home healthcare, industrial safety monitoring, and remote applications.

The existing systems have several limitations such as:

- High cost of monitoring equipment
- Lack of portability
- Limited wireless communication
- Manual monitoring requirements
- Difficulty in continuous real-time observation
- Limited remote healthcare support
- Lack of environmental dust monitoring and asthma risk prediction

Due to these limitations, there is a need for an advanced IoT-based healthcare and dust prediction monitoring system that can provide portable, wireless, and real-time user monitoring efficiently while also detecting harmful dust conditions and predicting asthma-related health risks.

3.4 EXISTING SYSTEM OVERVIEW

The proposed system is an IoT-based real-time healthcare and dust risk prediction monitoring system developed using ESP32 and Raspberry Pi for monitoring user vital signs continuously along with environmental dust conditions.

The system is mainly designed to measure important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and dust concentration in real time and display the collected information through a monitoring station. The system also predicts possible asthma risks caused due to harmful dust exposure.

The system consists of two main sections: the portable health monitoring unit and the monitoring station. The portable unit includes the ESP32 microcontroller, MAX30100 pulse oximeter sensor, DS18B20 temperature sensor, dust sensor, rechargeable battery, and TP4056 charging module.

The MAX30100 sensor is used to measure heart rate and oxygen saturation level, while the DS18B20 sensor is used to measure body temperature. The dust sensor is additionally used for monitoring surrounding dust levels and supporting instant asthma risk prediction.

The ESP32 acts as the main processing and communication unit of the system. It collects the sensor readings, processes the data, predicts possible dust-related asthma conditions, and transmits the information wirelessly through Wi-Fi communication.

The Raspberry Pi functions as the monitoring station that receives the transmitted data from the ESP32 and displays the live health parameters and dust monitoring information on a TFT display screen for continuous observation. The overall system is designed to provide portable, low-cost, wireless, and real-time healthcare and dust prediction monitoring for hospitals, home healthcare, industries, and remote user monitoring applications.

The project also supports future enhancements such as cloud connectivity, mobile application integration, emergency alert systems, dust analytics, and AI-based asthma risk prediction for advanced healthcare solutions.

3.5 TECHNOLOGIES USED

ESP32 WROOM Development Board

The ESP32 is used as the main microcontroller and Wi-Fi communication module in the project. It collects sensor data, dust monitoring information, processes the readings, predicts possible asthma risks, and transmits the information wirelessly to the Raspberry Pi.

Raspberry Pi 4 Model B

The Raspberry Pi acts as the monitoring station and server for receiving and displaying user health data and dust monitoring data in real time. It controls the TFT display and processes incoming data from the ESP32.

MAX30100 Pulse Oximeter Sensor

The MAX30100 sensor is used for measuring heart rate and oxygen saturation level (SpO2). It works using infrared light sensing technology to detect pulse and oxygen levels in the blood.

TP4056 Charging Module and 18650 Battery

The TP4056 module and rechargeable 18650 battery are used to provide portable power supply and battery charging support for the ESP32 monitoring unit.

Dust Sensor

The dust sensor is used for detecting environmental dust concentration levels in real time. The collected dust data is used for predicting possible asthma risk conditions caused due to harmful dust exposure.

3.5 Inch TFT Display

The TFT display is connected to the Raspberry Pi for visualizing real-time user health parameters, dust monitoring values, asthma risk status, heart rate, SpO2, and temperature.

DS18B20 Temperature Sensor

The DS18B20 digital sensor is used for measuring body temperature accurately and transmitting the readings to the ESP32.

Software Technologies

Arduino IDE

Arduino IDE is used for programming the ESP32 microcontroller and interfacing the sensors and dust prediction modules with the system.

Raspberry Pi OS

Raspberry Pi OS is used as the operating system for running the monitoring applications on Raspberry Pi.

Python Programming Language

Python is used in the Raspberry Pi for receiving sensor data, processing information, dust analysis, asthma prediction monitoring, and displaying live readings on the TFT display.

Thonny IDE

Thonny IDE is used for developing and testing Python programs in the Raspberry Pi environment.

Wi-Fi Communication

Wi-Fi technology is used for wireless data transmission between the ESP32 and Raspberry Pi monitoring station in real time.

3.6 WORKING OF EXISTING SYSTEM

The existing healthcare monitoring system mainly works through traditional medical devices that are used to measure user vital signs separately. In hospitals and healthcare centers, healthcare professionals manually monitor parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and respiratory conditions using individual monitoring devices. The measured values are observed directly from the devices and recorded manually for user analysis. Existing systems generally do not provide continuous dust monitoring and instant asthma risk prediction capabilities.

Most existing systems are designed for hospital-based monitoring and require continuous supervision by medical staff. The devices are generally connected through wired systems and are not suitable for portable or remote monitoring applications.

In many cases, the monitoring process is limited to a specific location, and continuous real-time observation becomes difficult for users outside hospital environments. The existing monitoring systems usually involve separate instruments for measuring different health parameters and dust conditions. The collected data is not always transmitted wirelessly, and remote access to user information and environmental monitoring status is limited. Some advanced systems provide digital monitoring, but they are often expensive and require complex infrastructure.

The overall working of the existing system mainly depends on manual observation, wired communication, and stationary monitoring equipment, which reduces flexibility and portability in healthcare monitoring applications and environmental dust analysis.Due to these limitations, there is a need for an IoT-based real-time monitoring system that provides wireless communication, portability, continuous healthcare observation, dust monitoring, and asthma risk prediction efficiently.

Disadvantages of Existing System

- Traditional healthcare monitoring systems require continuous manual observation by medical staff.

- Most existing systems are expensive and difficult to afford for small hospitals and home healthcare applications.
- The monitoring devices are usually bulky and not suitable for portable use.
- Existing systems mainly depend on wired communication, which reduces flexibility and mobility.
- Continuous real-time monitoring is difficult in remote and rural areas.
- Different health parameters are often measured using separate devices, increasing system complexity.
- Remote monitoring and wireless data transmission are limited in many conventional systems.
- Existing systems may require complex infrastructure and maintenance.
- Manual recording of user data can increase the chances of human error.
- Lack of portability and automation reduces the efficiency of user healthcare monitoring.
- Existing systems provide limited support for dust monitoring and instant asthma risk prediction.

3.7 PROPOSED SYSTEM

The proposed system is an IoT-based real-time healthcare and dust risk prediction monitoring system developed using ESP32 and Raspberry Pi for continuous monitoring of user vital signs and environmental dust conditions.

The system is designed to measure important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and dust concentration using integrated sensors and wireless communication technology. The system also predicts possible asthma risk conditions caused due to harmful dust exposure.

In the proposed system, the MAX30100 sensor is used to measure heart rate and SpO2 levels, while the DS18B20 sensor is used to measure body temperature. A dust sensor is additionally used for monitoring environmental dust concentration levels and predicting asthma-related breathing risks.

These sensors are connected to the ESP32 microcontroller, which acts as the main processing and communication unit. The ESP32 collects the sensor readings, processes the data, predicts possible asthma risk conditions, and transmits the information wirelessly through Wi-Fi communication.

The Raspberry Pi acts as the monitoring station that receives the transmitted data from the ESP32 and displays the live health parameters and dust monitoring information on a TFT display screen in real time. The system provides continuous monitoring and visualization of user health conditions and environmental safety for easy observation and analysis.

The proposed system is designed to be portable, rechargeable, low-cost, and easy to use. The ESP32 portable unit is powered using an 18650 battery and TP4056 charging module, making the device suitable for wearable and remote healthcare applications. The system reduces manual monitoring efforts and supports wireless real-time healthcare and dust monitoring efficiently.

The proposed system also provides opportunities for future enhancements such as cloud connectivity, mobile application integration, emergency alert systems, dust analytics, GPS tracking, and AI-based asthma risk prediction for advanced healthcare and environmental monitoring solutions.

3.8 SYSTEM OVERVIEW

In our project, we developed an IoT-based real-time healthcare and dust risk prediction monitoring system using ESP32 and Raspberry Pi for continuous monitoring of user vital signs and environmental dust conditions. The system is mainly designed to measure important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and dust concentration in real time using sensor technology and wireless communication. The system also predicts possible asthma risk conditions caused due to harmful dust exposure.

Our system consists of two main sections: the portable health monitoring unit and the monitoring station. The portable unit contains the ESP32 microcontroller, MAX30100 pulse oximeter sensor, DS18B20 temperature sensor, dust sensor, rechargeable battery, and TP4056 charging module. The MAX30100 sensor is used to measure heart rate and SpO2 levels, while the DS18B20 sensor is used to measure body temperature. In our system, the ESP32 acts as the main processing and communication unit. It collects the sensor data, dust concentration values, processes the information, predicts possible asthma risks, and transmits the information wirelessly through Wi-Fi communication.

The Raspberry Pi acts as the monitoring station that receives the transmitted data from the ESP32 and displays the live health parameters and dust monitoring information on a TFT display screen for continuous monitoring and analysis.

The entire system is designed to provide portable, low-cost, wireless, and real-time healthcare and dust prediction monitoring for hospitals, home healthcare, industries, and remote user monitoring applications where continuous observation of user health and environmental safety is required.

The project also supports future enhancements such as cloud connectivity, mobile application support, emergency alert systems, dust analytics, and AI-based asthma risk prediction for advanced healthcare solutions.

3.9 TECHNOLOGIES USED

In our project, we used both hardware and software technologies to develop an IoT-based real-time healthcare monitoring and dust risk prediction system using ESP32 and Raspberry Pi. These technologies help in collecting, processing, transmitting, analyzing, and visualizing user health data and environmental dust levels in real time.

Hardware Technologies

ESP32 WROOM Development Board

We used the ESP32 microcontroller as the main processing and communication unit of the system. It collects sensor data, dust monitoring values, processes the readings, predicts possible asthma risks, and transmits the information wirelessly through Wi-Fi communication.

Raspberry Pi 4 Model B

We used Raspberry Pi as the monitoring station for receiving and displaying real-time user health data and environmental dust monitoring information. It also controls the TFT display and processes incoming sensor information.

MAX30100 Pulse Oximeter Sensor

The MAX30100 sensor is used in our project for measuring heart rate and oxygen saturation level (SpO2). It works using infrared light sensing technology.

DS18B20 Temperature Sensor

We used the DS18B20 digital sensor for measuring body temperature accurately and transmitting the readings to the ESP32.

Dust Sensor

The dust sensor is used for monitoring environmental dust concentration levels in real time. The collected data is used for predicting asthma risk conditions caused due to dust exposure.

3.5 Inch TFT Display

The TFT display is used for visualizing live user health parameters such as heart rate, SpO2, body temperature, dust levels, and asthma risk status on the Raspberry Pi monitoring station.

TP4056 Charging Module and 18650 Battery

We used the TP4056 charging module and rechargeable 18650 battery to provide portable power supply and battery charging support for the ESP32 monitoring unit.

Software Technologies

Arduino IDE

We used Arduino IDE for programming the ESP32 microcontroller and interfacing the sensors and dust prediction modules with the system.

Raspberry Pi OS

Raspberry Pi OS is used as the operating system for running the monitoring applications on Raspberry Pi.

Python Programming Language

We used Python programming language in Raspberry Pi for receiving sensor data, processing information, dust analysis, asthma prediction monitoring, and displaying live readings on the TFT display.

Thonny IDE

Thonny IDE is used for writing, testing, and executing Python programs in the Raspberry Pi environment.

Wi-Fi Communication

Wi-Fi technology is used in our project for wireless data transmission between ESP32 and Raspberry Pi in real time.

3.10 WORKING OF PROPOSED SYSTEM

In our proposed system, the healthcare monitoring process begins with the collection of user vital signs and environmental dust levels using sensors connected to the ESP32 microcontroller.

The MAX30100 pulse oximeter sensor is used to measure heart rate and oxygen saturation level (SpO2), while the DS18B20 sensor is used to measure body temperature. A dust sensor is additionally used to monitor environmental dust concentration and predict asthma risk conditions caused due to harmful dust exposure. These sensors continuously collect health and environmental data from the user in real time.

The ESP32 acts as the main processing and communication unit of the system. It reads the sensor values, processes the collected data, converts the readings into understandable health parameters such as heart rate in beats per minute (BPM), oxygen saturation percentage, body temperature in degree Celsius, and dust concentration levels. The ESP32 also predicts possible asthma-related breathing risks based on dust exposure levels.

After processing the data, the ESP32 connects to the Wi-Fi network and transmits the sensor readings and dust prediction information wirelessly to the Raspberry Pi monitoring station. Wireless communication is achieved using IoT technology, which enables real-time data transfer between the portable monitoring unit and the monitoring station.

The Raspberry Pi receives the transmitted health data and dust monitoring information from the ESP32 and processes the information for visualization. The received values are displayed on a 3.5-inch TFT display screen connected to the Raspberry Pi.

The display continuously shows live user parameters such as heart rate, SpO2 level, body temperature, dust concentration, asthma risk status, and system status for easy monitoring.

The entire system works continuously in real time, allowing healthcare providers or users to observe user health conditions instantly. The portable battery-powered design of the ESP32 unit makes the system compact, rechargeable, and suitable for hospitals, home healthcare, industries, and remote user monitoring applications.

Advantages

- The proposed system provides real-time monitoring of user vital signs continuously.
- The system supports wireless communication using Wi-Fi technology for easy data transmission.
- The portable design makes the device suitable for wearable and remote healthcare applications.
- The system is low-cost compared to traditional healthcare monitoring equipment.
- The ESP32 and Raspberry Pi combination provides efficient data processing and monitoring.
- The system reduces manual monitoring efforts in hospitals and healthcare centers.
- Real-time visualization of health parameters and dust monitoring improves user observation and analysis.
- The rechargeable battery system makes the portable unit easy to use and power efficient.
- The system is compact, lightweight, and easy to operate.
- It can be used in hospitals, home healthcare, industries, and remote user monitoring applications.
- The project supports future enhancements such as cloud storage, mobile applications, emergency alerts, dust analytics, and AI-based asthma risk prediction.
- The system helps in early detection of abnormal health conditions and dust-related asthma risks through continuous monitoring.

3.11 FEASIBILITY STUDY

The feasibility study of our project evaluates the practicality and effectiveness of developing the Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor using ESP32 and Raspberry Pi. The study helps in understanding whether the proposed system can be implemented successfully in terms of technical requirements, cost, operation, environmental monitoring capability, and future usability.

3.11.1 Technical Feasibility

The proposed system is technically feasible because the required hardware and software components are easily available and compatible with each other. The ESP32 microcontroller supports Wi-Fi communication and sensor interfacing, while the Raspberry Pi provides efficient processing, dust analysis, asthma risk prediction, and real-time data visualization capabilities.

Sensors such as MAX30100, DS18B20, and dust sensors can accurately measure vital signs and environmental conditions and communicate effectively with the ESP32. The software tools including Arduino IDE, Python, and Raspberry Pi OS are open-source and suitable for IoT application development.

3.11.2 Economic Feasibility

The project is economically feasible because the overall cost of the system is comparatively low when compared to traditional healthcare monitoring systems. The hardware components used in the project are affordable, compact, and easily available in the market. Since the project uses open-source software platforms, additional software licensing costs are avoided. The low-cost design makes the system suitable for hospitals, clinics, industries, and home healthcare applications.

3.11.3 Operational Feasibility

The proposed system is simple to operate and user-friendly. The healthcare parameters and dust concentration levels are automatically collected, processed, and displayed in real time without requiring complex manual operations. Wireless communication between ESP32 and Raspberry Pi improves system flexibility and portability. The system can be easily used by healthcare professionals and users for continuous monitoring purposes.

3.11.4 Practical Feasibility

The system is practically feasible for real-time healthcare monitoring and dust risk prediction applications because it provides portable and continuous observation of user vital signs and environmental safety. The rechargeable battery-powered design allows the monitoring unit to be carried easily for remote healthcare applications. The project can be implemented in hospitals, home healthcare environments, industries, and remote monitoring systems effectively.

3.11.5 Future Feasibility

The project also supports future enhancements such as cloud storage, mobile application integration, emergency alert systems, GPS tracking, dust analytics, and AI-based asthma risk prediction. These features can improve the overall functionality and efficiency of the healthcare and environmental monitoring system in the future.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 INTRODUCTION

System requirements define the hardware and software environment necessary for the development, implementation, and maintenance of the proposed Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor using ESP32 and Raspberry Pi. This chapter describes the minimum hardware specifications, software tools, communication technologies, and development platforms required for implementing the proposed healthcare monitoring, dust monitoring, and asthma risk prediction system. The system is designed to provide real-time monitoring of user vital signs such as heart rate, oxygen saturation level (SpO2), body temperature, and environmental dust concentration through wireless communication and live data visualization.

4.2 HARDWARE REQUIREMENTS

The proposed system requires hardware components for sensing, processing, wireless communication, dust analysis, asthma prediction, and real-time visualization of user health parameters.

- ESP32 WROOM Development Board is used as the main microcontroller for sensor interfacing and Wi-Fi communication.
- Raspberry Pi 4 Model B acts as the monitoring station for receiving and displaying health data and dust monitoring information in real time.
- The MAX30100 pulse oximeter sensor is used for measuring heart rate and oxygen saturation level (SpO2), while the DS18B20 digital sensor is used for measuring body temperature.
- A dust sensor is connected to the Raspberry Pi or ESP32 for live visualization of dust concentration levels and asthma risk prediction.

- The ESP32 portable unit is powered using a rechargeable 18650 battery along with a TP4056 charging module for efficient power management.
- A stable Wi-Fi network connection is required for wireless communication between ESP32 and Raspberry Pi.
- Additional basic components such as jumper wires, breadboard, USB cables, and power adapters are also required for proper system setup and operation.

4.3 SOFTWARE REQUIREMENTS

The software requirements include development tools, programming platforms, and runtime technologies used for implementing the healthcare monitoring, dust monitoring, and asthma risk prediction system.

- Arduino IDE is used for programming the ESP32 microcontroller and interfacing the sensors and dust prediction modules with the system.
- Raspberry Pi OS is used as the operating system for running the monitoring applications on Raspberry Pi.
- Python programming language is used in Raspberry Pi for receiving sensor data, processing information, dust analysis, asthma risk prediction, and displaying live health parameters on the TFT display.
- Thonny IDE is used for writing, testing, and executing Python programs in the Raspberry Pi environment.
- Wi-Fi communication technology is used for wireless data transmission between ESP32 and Raspberry Pi. The software environment is designed to support real-time data acquisition, processing, visualization, environmental dust monitoring, and continuous monitoring of user health conditions.

4.4 SOFTWARE DESCRIPTION

The proposed healthcare monitoring system is an IoT-based real-time monitoring platform developed using ESP32 and Raspberry Pi technologies. The system integrates sensor interfacing, wireless communication, dust processing, asthma risk prediction, data processing, and real-time visualization for monitoring user vital signs continuously.

The MAX30100 sensor is used to detect heart rate and oxygen saturation level (SpO2) using infrared sensing technology, while the DS18B20 sensor measures body temperature accurately. A dust sensor is additionally used to monitor surrounding dust concentration levels and predict asthma-related risks caused by harmful environmental conditions.

The ESP32 microcontroller collects the sensor readings, processes the data, predicts possible asthma risk conditions, and transmits the information wirelessly through Wi-Fi communication.

The Raspberry Pi monitoring station receives the transmitted data and displays the live health parameters, dust monitoring information, and asthma risk status on a TFT display screen. The system provides continuous observation of user health conditions and environmental safety in real time. The portable and rechargeable design of the ESP32 unit makes the system suitable for hospitals, home healthcare, industrial safety monitoring, and remote user monitoring applications. The software implementation ensures reliable communication, smooth system performance, efficient dust prediction, and effective real-time healthcare monitoring.

CHAPTER 5

SYSTEM DESIGN

5.1 INTRODUCTION

The proposed system follows an IoT-based healthcare monitoring architecture developed using ESP32 and Raspberry Pi. The architecture is divided into multiple sections including Sensor Layer, Processing and Communication Layer, Power Management Layer, Monitoring Layer, Dust Analysis Layer, and Prediction Layer, ensuring continuous real-time monitoring, environmental dust analysis, asthma prediction, and wireless communication of user health parameters.

5.1.1 Sensor Layer

The Sensor Layer is responsible for collecting user vital signs and environmental dust conditions in real time. It includes the MAX30100 pulse oximeter sensor, DS18B20 temperature sensor, and dust sensor connected to the ESP32 microcontroller.

The MAX30100 sensor is used to measure heart rate and oxygen saturation level (SpO2) using the I2C communication protocol. The DS18B20 digital sensor measures body temperature using the 1-Wire communication protocol. The dust sensor continuously monitors surrounding dust concentration levels for asthma risk prediction. These sensors continuously collect health and environmental data from the user and send the readings to the ESP32 for processing.

5.1.2 Processing and Communication Layer

The Processing and Communication Layer is handled by the ESP32 microcontroller. This layer is responsible for processing sensor readings, dust concentration levels, asthma prediction analysis, converting the values into understandable health parameters, and transmitting the data wirelessly. The ESP32 processes the collected data and sends it to the Raspberry Pi monitoring station using Wi-Fi or MQTT communication. This layer ensures reliable wireless communication, continuous dust analysis, asthma prediction, and real-time data transfer between the portable unit and monitoring station.

5.1.3 Power Management Layer

The Power Management Layer provides power supply and charging support for the portable monitoring unit. It consists of a rechargeable 18650 battery and TP4056 charging module.

The TP4056 module is used for battery charging and power regulation, while the 18650 battery powers the ESP32 and sensor modules. This layer makes the system portable, rechargeable, and suitable for remote healthcare, dust monitoring, and asthma prediction applications.

5.1.4 Monitoring Layer

The Monitoring Layer is implemented using Raspberry Pi 4, which acts as the monitoring server and data visualization unit. The Raspberry Pi receives the transmitted health data and dust monitoring information from the ESP32 and processes the information for continuous user monitoring and asthma prediction visualization.

The received health parameters, dust concentration levels, and asthma risk status are displayed on a TFT display screen for real-time observation. This layer allows healthcare providers and users to monitor user conditions and environmental safety continuously and efficiently.

5.1.5 Architecture Flow

The workflow of the proposed system is as follows:

User & Environment → MAX30100 + DS18B20 + Dust Sensors → ESP32 Processing Unit → Wi-Fi / MQTT Communication → Raspberry Pi Monitoring Server → TFT Display Output → Real-Time User Monitoring & Asthma Risk Prediction

5.2 SYSTEM ARCHITECTURE

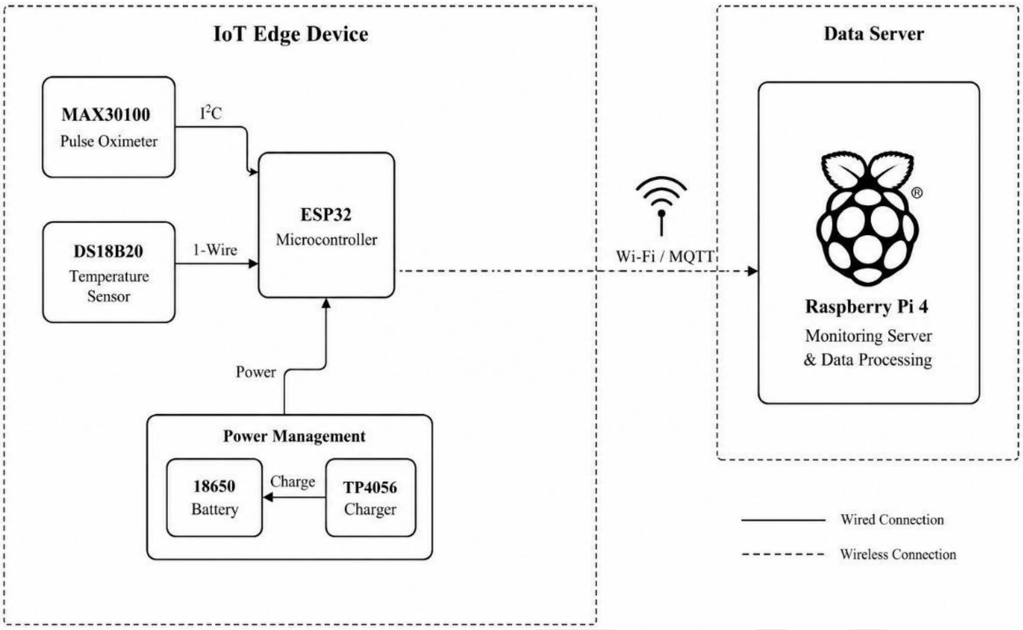


Fig 5.2 System Architecture

The proposed IoT-Based Real-Time Lung Health Monitoring System consists of two main sections: the IoT edge device and the monitoring server. The architecture is designed to provide continuous monitoring of user vital signs through wireless communication and real-time data visualization.

In the IoT edge device section, the MAX30100 pulse oximeter sensor and DS18B20 temperature sensor are connected to the ESP32 microcontroller. The MAX30100 sensor communicates with ESP32 using the I2C protocol for measuring heart rate and oxygen saturation level (SpO2), while the DS18B20 sensor uses the 1-Wire communication protocol for measuring body temperature.

The ESP32 acts as the main processing and communication unit of the system. It collects sensor data, processes the readings, and transmits the health parameters wirelessly through Wi-Fi or MQTT communication to the Raspberry Pi monitoring station. A buzzer module is also connected to the ESP32 through GPIO pins to provide alert notifications during abnormal health conditions.

The portable monitoring unit is powered using a rechargeable 18650 battery and TP4056 charging module for efficient power management and portability. The Raspberry Pi 4 acts as the data server and monitoring station that receives the transmitted sensor data and displays the live health parameters for continuous user monitoring.

The overall architecture provides a low-cost, portable, wireless, and real-time healthcare monitoring solution suitable for hospitals, home healthcare, and remote user monitoring applications.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 MODULES OF THE PROJECT

The proposed Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor consists of multiple modules, where each module is responsible for performing specific healthcare monitoring, environmental dust monitoring, and asthma prediction operations.

The modular design improves system efficiency, portability, wireless communication, environmental analysis, and real-time user monitoring performance.

The main modules included in the project are Sensor Module, ESP32 Processing Module, Wireless Communication Module, Raspberry Pi Monitoring Module, Dust Monitoring Module, Prediction Module, Display Module, and Power Management Module. Each module works together to provide continuous monitoring of user vital signs such as heart rate, oxygen saturation level (SpO2), body temperature, and environmental dust concentration.

6.2 SENSOR MODULE

This module is responsible for collecting user vital signs and environmental dust levels in real time. The MAX30100 pulse oximeter sensor is used for measuring heart rate and oxygen saturation level (SpO2), while the DS18B20 digital sensor is used for measuring body temperature. A dust sensor is additionally used for monitoring environmental dust concentration levels and supporting asthma risk prediction.

The sensors continuously collect health and environmental data from the user and surroundings and transmit the readings to the ESP32 microcontroller for processing. The MAX30100 sensor communicates using the I2C protocol, while the DS18B20 sensor uses 1-Wire communication.

6.3 ESP32 PROCESSING MODULE

The ESP32 module acts as the main processing and control unit of the system. It receives sensor readings and dust concentration values, processes the data, predicts possible asthma risks, and converts the information into readable health parameters such as BPM, SpO2 percentage, body temperature, and dust level status.

The ESP32 also manages wireless communication and controls the overall functioning of the portable healthcare and dust prediction monitoring device efficiently

6.4 WIRELESS COMMUNICATION MODULE

This module handles wireless data transmission between the ESP32 and Raspberry Pi monitoring station. Wi-Fi or MQTT communication technology is used for transmitting health data, dust concentration values, and asthma prediction information in real time.

The wireless communication module ensures smooth and continuous transfer of user vital sign data and environmental monitoring information without complex wired connections, making the system suitable for portable and remote healthcare applications.

6.5 RASPBERRY PI MONITORING MODULE

The Raspberry Pi module acts as the monitoring server and data processing unit of the system. It receives the transmitted health data and dust monitoring information from the ESP32 and processes the information for real-time monitoring and asthma prediction visualization.

The Raspberry Pi continuously monitors incoming sensor values and supports live visualization of user health conditions, dust concentration levels, and asthma risk status through the connected TFT display.

6.6 DISPLAY MODULE

This module is responsible for displaying live user health parameters and environmental monitoring information on the TFT display screen connected to the Raspberry Pi. The display shows important information such as heart rate, oxygen saturation level (SpO2), body temperature, dust concentration levels, asthma prediction status, and system status.

The display module allows healthcare providers and users to observe user conditions and environmental safety continuously and efficiently in real time.

6.7 POWER MANAGEMENT MODULE

This module provides power supply and charging support for the portable monitoring unit. It consists of a rechargeable 18650 battery and TP4056 charging module.

The TP4056 charging module manages battery charging and power regulation, while the 18650 battery powers the ESP32 and sensor modules. This module makes the healthcare and dust monitoring device portable, rechargeable, and suitable for continuous operation in real-time healthcare and environmental monitoring applications.

CHAPTER 7

EXPERIMENTAL RESULTS

7.1 INTRODUCTION

This chapter presents the results obtained from the implementation of the Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor using ESP32 and Raspberry Pi. The developed system includes important components such as biomedical sensors, dust sensing modules, wireless communication modules, Raspberry Pi monitoring station, and real-time data visualization through a TFT display. The system successfully performs continuous monitoring of user vital signs including heart rate, oxygen saturation level (SpO2), body temperature, and environmental dust concentration for instant asthma risk prediction.

The ESP32 microcontroller collects health data from the MAX30100 pulse oximeter sensor, DS18B20 temperature sensor, and dust sensor and transmits the information wirelessly to the Raspberry Pi using Wi-Fi or MQTT communication. The Raspberry Pi monitoring station receives the transmitted data and displays the live readings continuously on the TFT display screen along with dust monitoring information and asthma prediction status.

Users are able to monitor user health conditions and environmental dust exposure in real time through the monitoring interface. The portable ESP32 monitoring unit powered by a rechargeable battery provides continuous healthcare and dust monitoring without complex wired connections. The wireless communication between ESP32 and Raspberry Pi ensures smooth real-time data transmission, dust analysis, and asthma prediction performance.

Compared to traditional healthcare monitoring methods, the proposed system provides a low-cost, portable, wireless, and efficient healthcare and dust risk prediction solution for hospitals, home healthcare, industries, and remote user monitoring applications. The experimental results demonstrate stable sensor performance, reliable wireless communication, environmental dust monitoring, real-time visualization, and effective user health monitoring using IoT technology.

7.2 SERVER SETUP OUTPUT

The Raspberry Pi monitoring station was successfully configured as a local Flask server for receiving real-time health data and dust monitoring information from the ESP32 through Wi-Fi communication. The server application was executed using Python on Raspberry Pi OS through the terminal interface displayed on the TFT screen.

The Flask server was started using the following command:

python3 server.py

After execution, the server initialized successfully and started listening for incoming connections from the ESP32 device. The terminal output confirmed that the Flask application was running on all available network interfaces using port 5000.

The server generated both local and network IP addresses for communication:

Running on http://127.0.0.1:5000
Running on http://172.29.43.149:5000

The IP address was later used by the ESP32 portable monitoring unit to transmit user vital sign data such as heart rate, oxygen saturation level (SpO2), body temperature, dust concentration values, and asthma prediction information to the Raspberry Pi monitoring station in real time.

The successful server setup verified that the Raspberry Pi was capable of receiving wireless health data, dust monitoring information, and asthma prediction results continuously and processing the information for real-time healthcare and environmental monitoring applications.

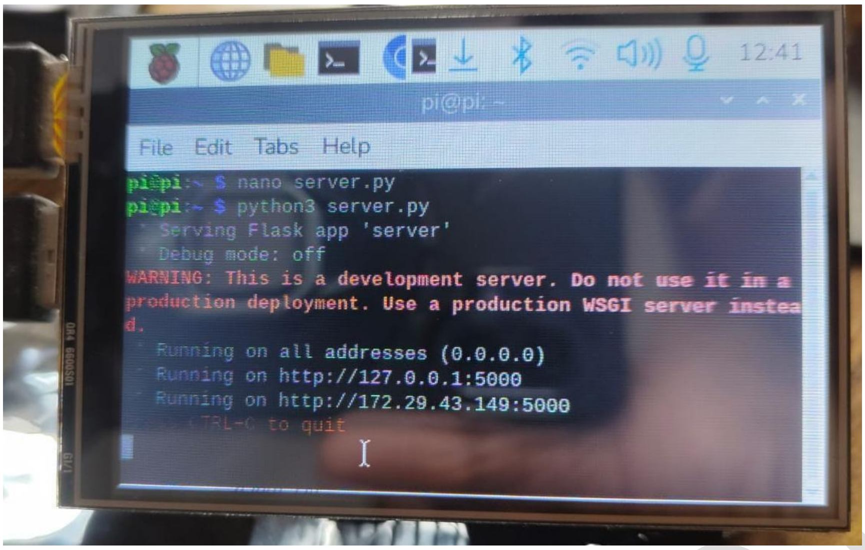


Fig 7.2 Server Setup Output

7.3 ESP32 CONNECTION SETUP WITH RASPBERRY PI

The ESP32 microcontroller was configured to establish wireless communication with the Raspberry Pi monitoring server using Wi-Fi and HTTP communication protocol. The ESP32 program was developed and uploaded using Arduino IDE.

The ESP32 first connects to the local Wi-Fi network and then sends HTTP requests containing sensor data and dust monitoring information to the Raspberry Pi Flask server. The Raspberry Pi server IP address and port number were specified in the ESP32 program for successful communication.

The following server URL was used in the ESP32 code:

String serverName = "http://172.29.43.149:5000/data?msg=HELLO_PI";

The HTTP client library initializes the connection using:

http.begin(serverName);

The ESP32 sends a GET request to the Raspberry Pi server using:

int httpResponseCode = http.GET();

The Serial Monitor output in Arduino IDE confirms successful Wi-Fi connection and communication between ESP32 and Raspberry Pi. Initially, the ESP32 attempted to connect to Wi-Fi repeatedly until the connection was established successfully.

The output displayed:

Connecting to WiFi...
WiFi Connected
HTTP Response code: 200

The HTTP response code 200 indicates that the Raspberry Pi server successfully received the request sent from the ESP32 device. This confirms proper wireless communication between the portable monitoring unit and the Raspberry Pi monitoring station for transmitting healthcare data, dust monitoring values, and asthma prediction information.

The successful ESP32 connection setup enabled real-time transmission of user health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, environmental dust concentration, and asthma prediction status from the sensors to the Raspberry Pi server for continuous healthcare and environmental monitoring.

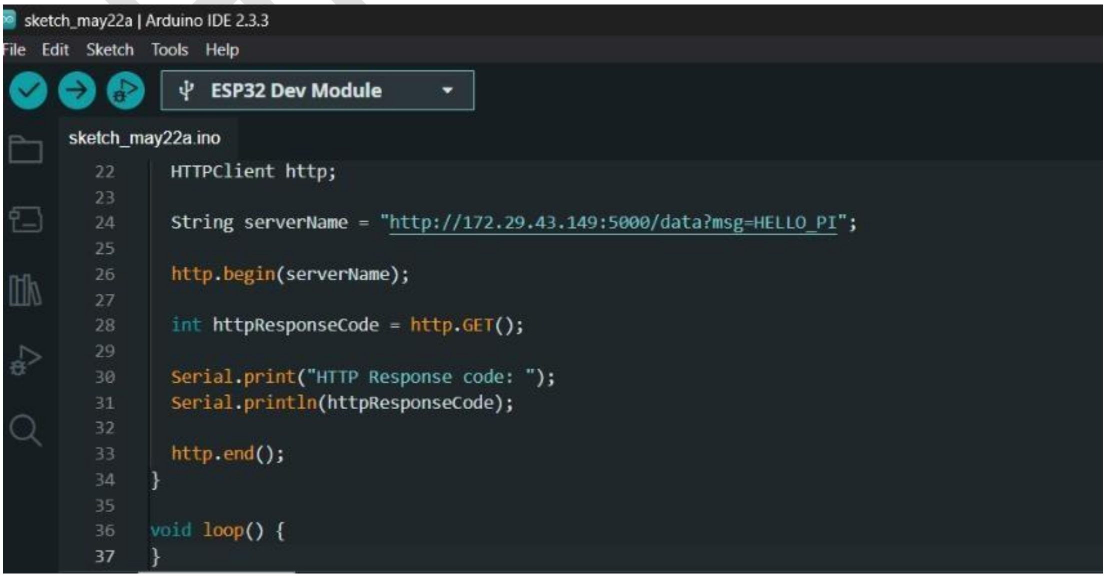
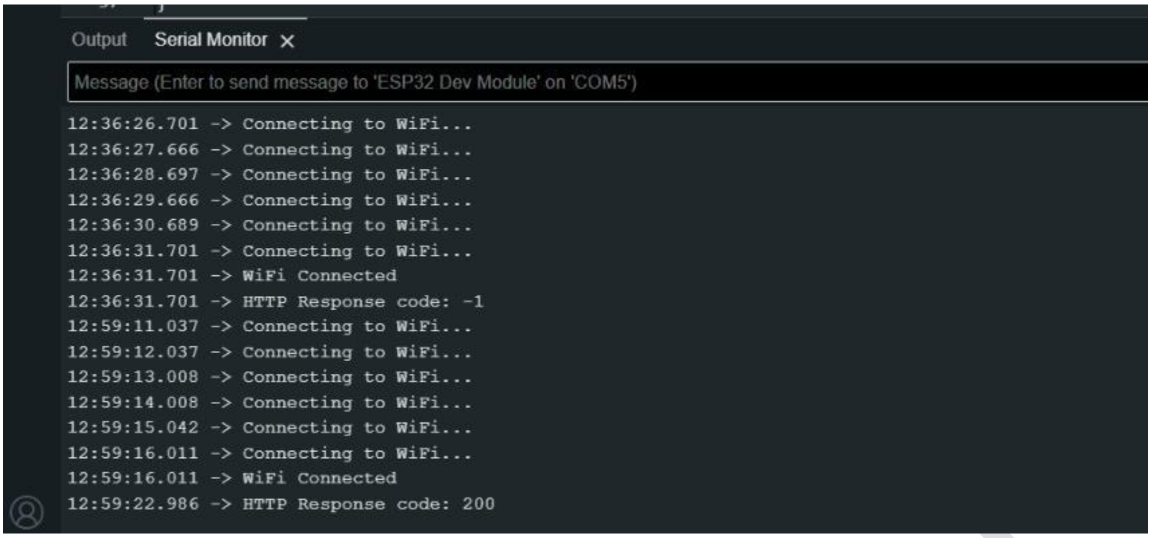


Fig 7.3 Esp32 Connection Setup With Raspberry Pi



7.4 FINAL HARDWARE SETUP

The final implementation of the proposed Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor was successfully developed using ESP32 and Raspberry Pi for real-time healthcare monitoring and environmental dust analysis. The hardware setup consists of the ESP32 portable monitoring unit, biomedical sensors, dust sensor, rechargeable power supply, and Raspberry Pi monitoring station with TFT display.

The ESP32 development board was connected with the MAX30100 pulse oximeter sensor, DS18B20 temperature sensor, and dust sensor for measuring heart rate, oxygen saturation level (SpO2), body temperature, and dust concentration levels. The system was assembled on a PCB board for stable hardware integration and efficient connectivity between the modules.

A rechargeable 18650 battery and TP4056 charging module were used to provide portable power supply for the ESP32 monitoring unit. The ESP32 collected sensor readings and dust values continuously and transmitted the health data and asthma prediction information wirelessly to the Raspberry Pi monitoring station using Wi-Fi communication.

The Raspberry Pi 4 with TFT display successfully received the transmitted sensor data and displayed the live health parameters, dust concentration values, and asthma prediction status in real time through a web-based monitoring interface. The display screen showed the measured temperature value of 37.0°C during testing, confirming successful communication between the ESP32 and Raspberry Pi systems.

The final setup demonstrated stable hardware operation, successful wireless data transmission, real-time dust analysis, asthma prediction functionality, and effective visualization of user health parameters. The implemented system proves that the proposed healthcare and dust prediction monitoring solution is portable, low-cost, rechargeable, and suitable for hospitals, home healthcare, industries, and remote user monitoring applications.

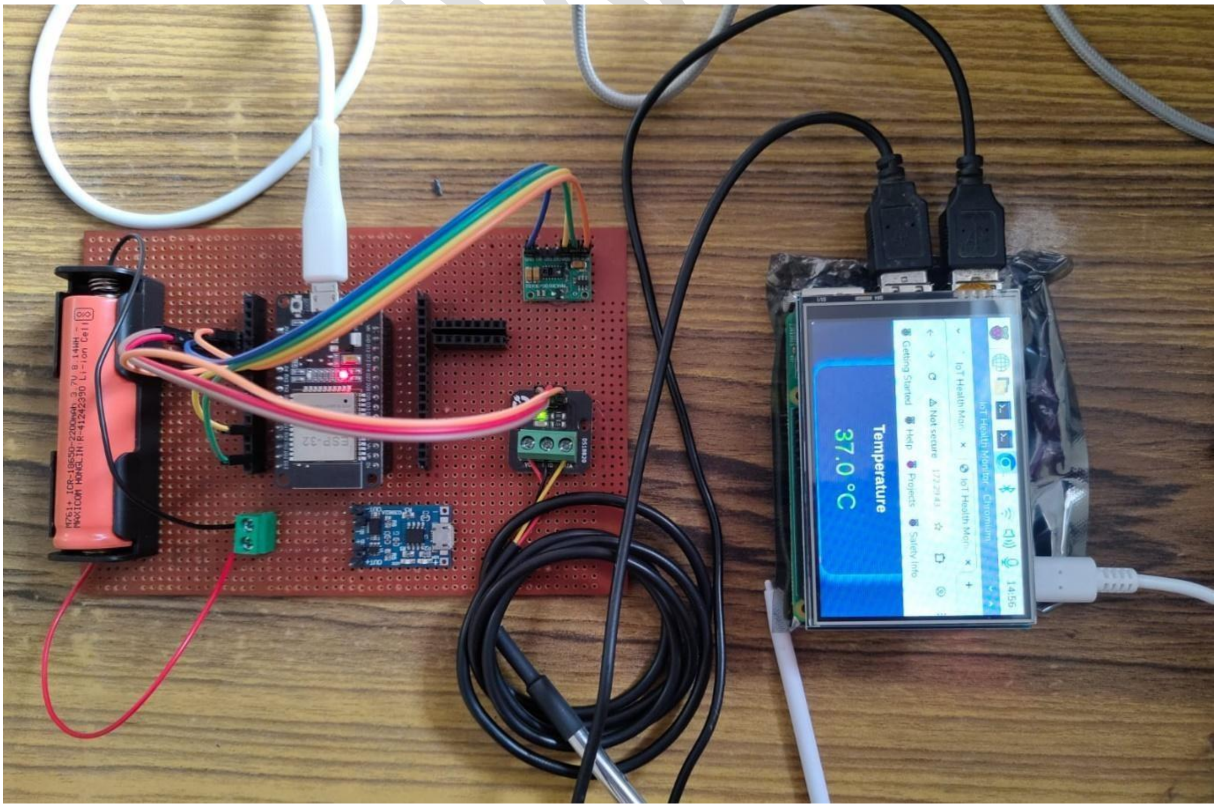


Fig 7.4 Final Hardware Setup

7.5 DISPLAY OUTPUT

The developed IoT Health Monitoring and Dust Prediction System successfully displayed real-time user vital signs through a web-based monitoring dashboard hosted on the Raspberry Pi server. The dashboard was accessed using the Raspberry Pi IP address through a web browser interface.

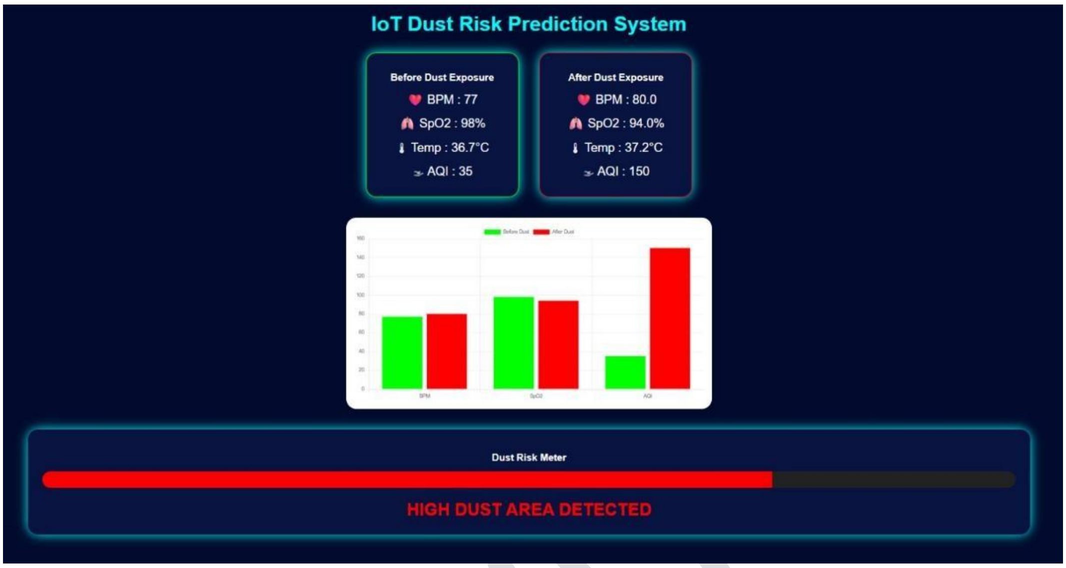
The monitoring interface displayed important healthcare parameters including heart rate, oxygen saturation level (SpO2), body temperature, environmental dust concentration, and user health status in real time. The ESP32 continuously transmitted sensor readings and dust monitoring values to the Raspberry Pi server using Wi-Fi communication, and the received data was updated dynamically on the dashboard.

During system testing, the displayed output values were:

- Heart Rate : 85 BPM
- SpO2 : 98%
- Temperature : 36.1°C
- Status : NORMAL

The dashboard interface was designed with a simple and user-friendly layout for easy observation of user health conditions and environmental safety. Different sections were used for displaying each health parameter and asthma prediction status clearly, enabling continuous real-time monitoring and quick medical analysis.

The successful output visualization confirms that the proposed system can effectively monitor, process, analyze dust exposure, predict asthma risks, and display user health data wirelessly using IoT technology. The system demonstrated stable communication between ESP32 and Raspberry Pi along with efficient real-time healthcare and environmental monitoring performance suitable for hospitals, home healthcare, industries, and remote user monitoring applications.



After dust prediction

Fig 7.5 Display Output

PARAMETERS	NORMAL RANGE	ABNORMAL CONDITION	STATUS
Heart Rate (BPM)	60 – 100 BPM	Below 60 or Above 100 BPM	Not Normal
SpO2 (Oxygen Level)	95% – 100%	Below 95%	Low Oxygen Level
Body Temperature	36.1°C – 37.2°C	Above 37.5°C or Below 35°C	Fever / Low Temperature
Dust Level	0 – 150 AQI	Above 150 AQI	Harmful Dust Condition
Asthma Risk Prediction	Low Dust + Normal SpO2	High Dust + Low SpO2	Asthma Risk Detected

CHAPTER 8

RESULTS AND FINDINGS

8.1 PERFORMANCE EVALUATION

The performance of the proposed Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor was evaluated based on sensor accuracy, wireless communication, dust monitoring capability, asthma prediction performance, real-time monitoring capability, and overall system performance. During testing, the system successfully collected user vital signs including heart rate, oxygen saturation level (SpO2), body temperature, and environmental dust concentration using the MAX30100, DS18B20, and dust sensors connected to the ESP32 microcontroller.

The ESP32 continuously processed the sensor readings and transmitted the health data, dust concentration values, and asthma prediction information wirelessly to the Raspberry Pi monitoring station using Wi-Fi communication. The communication between ESP32 and Raspberry Pi remained stable throughout the testing process, ensuring smooth and continuous real-time data transmission without significant delay.

The Raspberry Pi monitoring station successfully received and displayed the live user health parameters, environmental dust monitoring values, and asthma prediction status on the web-based dashboard and TFT display. The displayed values updated dynamically in real time, allowing continuous observation of user health conditions and environmental safety. The monitoring interface provided clear visualization of heart rate, SpO2 level, body temperature, dust levels, and asthma prediction status for easy analysis.

The portable monitoring unit powered using the rechargeable 18650 battery and TP4056 charging module functioned efficiently during continuous operation. The overall system performance remained stable with reliable sensor readings, accurate dust analysis, smooth wireless communication, and efficient asthma prediction monitoring.

Overall, the proposed system achieved effective real-time healthcare and environmental monitoring with good portability, low-cost implementation, stable wireless communication, efficient dust monitoring, and accurate asthma risk prediction. The results confirm that the system can be successfully used for hospitals, home healthcare, industries, and remote user monitoring applications using IoT technology.

CHAPTER 9

CONCLUSION AND FUTURE WORK

9.1 CONCLUSION

The developed Shrewd IoT Centred Instant Asthma due to Dust Risk Predictor using ESP32 and Raspberry Pi demonstrates an efficient and reliable solution for continuous user health monitoring, environmental dust analysis, and instant asthma risk prediction. The system successfully integrates biomedical sensors, dust sensing technology, wireless communication, real-time data processing, and live visualization for monitoring important health parameters such as heart rate, oxygen saturation level (SpO2), body temperature, and environmental dust concentration.

The proposed system continuously collects user vital signs and dust concentration values using the MAX30100, DS18B20, and dust sensors connected to the ESP32 microcontroller. The collected data is processed, analyzed for asthma risk conditions, and transmitted wirelessly through Wi-Fi communication to the Raspberry Pi monitoring station, where the live readings and asthma prediction status are displayed on a TFT screen for continuous observation.

The use of ESP32 and Raspberry Pi provides a low-cost, portable, and easy-to-use healthcare and environmental monitoring solution suitable for hospitals, home healthcare, industries, and remote user monitoring applications. The rechargeable battery-powered portable unit improves system mobility and allows real-time monitoring without complex wired connections.

Overall, the system achieves effective real-time healthcare monitoring, dust monitoring, and asthma prediction with improved portability, wireless communication, and continuous user observation. The project demonstrates strong potential for future healthcare and environmental monitoring applications by reducing manual monitoring efforts and supporting early detection of abnormal health conditions and dust-related asthma risks.

9.2 FUTURE WORK

- Mobile application integration for remote user monitoring through smartphones.
- Cloud storage implementation for maintaining user health records, dust monitoring history, and asthma prediction analysis.
- Emergency alert system for sending notifications during abnormal health conditions and dangerous dust exposure levels.
- AI-based healthcare and environmental analysis for predicting asthma risks and detecting respiratory diseases early.
- GPS tracking system for monitoring user location during emergencies.
- Addition of more biomedical sensors such as ECG, blood pressure, respiratory sensors, and advanced air quality sensors.
- Development of wearable and compact healthcare devices for improved portability.
- Integration of doctor and hospital management systems for advanced healthcare support.
- Improvement in wireless communication range and network security for safer data transmission.
- Real-time data analytics and graphical visualization for better healthcare, dust monitoring, and asthma prediction analysis.

REFERENCE

1. Hemaxi Vyas et al – "A Portable IoT-Based Health Monitoring Framework Using ESP32 for Isolated and Home-Based User Care" (2025)
2. Himanshi Manjhi – "IoT Based User Health Monitoring System Using ESP32 and Blynk App" (2025)
3. Jannatun Ferdous et al – "Implementation of IoT Based User Health Monitoring System Using ESP32 Web Server" (2023)
4. Suliman Abdulmalek et al – "IoT-Based Healthcare-Monitoring System towards Improving Quality of Life: A Review" (2022)
5. Yusra M. Obeidat et al – "An Embedded System Based on Raspberry Pi for Effective Electrocardiogram Monitoring" (2023)